

A design for tokenization in governmental investment

Tokenization in
governmental
investment

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Abstract

Purpose – The purpose of this study is to identify the possible efficiency gains in using tokenization for the execution of public expenditure on governmental investments.

Design/methodology/approach – Through design science research methodology, the exploratory research produced a tokenized prototype in the blockchain, through the Ernst and Young OpsChain traceability solution, allowing automated processes in the stages of public expense. A focus group composed of auditors from the public sector evaluated the possibility of improving the quality of information available in the audited entities, where the tokens created represent and register the actions of public agents in the blockchain Polygon.

Findings – The consensus of the experts in the focus group indicated that the use of tokenization could improve the quality of the information, since the possibility of recording the activities of public agents in the metadata of the tokens at each stage of the execution of the expenditure allows the audited entities the advantages of the information recorded on the blockchain, according to the following ranking: first the immutability of audited data, followed by reliability, transparency, accessibility and efficiency of data structures.

Originality/value – This research makes an empirical contribution to the real use of tokenization in blockchain technology to the public sector through a value chain in which tokens were created and moved between the wallets of public agents to represent, register and track the operations regarding public expense execution.

Keywords Blockchain tokenization, Public sector, Budget, Expense execution

Paper type Research paper

1. Introduction

Digital transformation and blockchain technology have been the subject of debate in the scientific, governmental and market media, notably by its association with native cryptocurrencies and the digital assets represented by tokens, which came to be called tokenization. Bitcoin was the first digital currency to disseminate this technology after the



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whitepaper r digital currency units have been issued on various platforms built in the blockchain format (Mohamed, 2021), many of them to facilitate international transactions and money movement and blockchain format (Mohamed, 2021), many of them to facilitate international transactions and money movement but also to perform other functions such as the representation of physical or intangible assets.

Technological advances and the expansion of the blockchain application domain have received tremendous attention from academia and industry, promising to change all aspects of digital business in the sector (Wang and Nixon, 2021).

Tokenization is one of blockchain's applications in the great race for the use of technology in diverse areas (Drasch *et al.*, 2020) and impacts other sectors of the economy and existing paradigms regarding the value, currencies, properties and thus control and tracking of assets. It is argued by scientists that almost all aspects of people's lives will be affected by the emergence and evolution of digitalization and peer-to-peer networks such as payments, savings, loans, insurance, identity cards, passports, diplomas, patents and employment contracts (Drescher, 2017).

In the public sector, blockchain technology has various possible or proven viable applications, particularly in voting and the possibility of transforming justice by combining transparency, collaboration and online citizen participation in solving society's major problems (Tapscott and Tapscott, 2016). As Li *et al.* (2019) report, blockchain is still under development and, like other new technologies, has faced some initial problems. However, many of the solutions for improving the effectiveness of blockchain applications use tokenization.

The tokens and coins generated by blockchain are regarded as cryptocurrency, a kind of encrypted digital or virtual currency (Li *et al.*, 2019). The tokens can represent all kinds of assets and be moved between digital wallets (inputs and outputs as debits and credits), as is done with the representative items of goods, rights and liabilities being moved in the form of a ledger, which may revolutionize accounting.

The tokenization process can be described as the encapsulation of value in tradeable units of account, called tokens or coins. The disruptive potential lies in expanding the concept of value that can be partitioned and traded beyond purely economic terms to include, for example, reputation, work, copyright, utility and voting rights. In other words, tokenization represents a form of digitalization of value, and just like the internet enabled the free and fast circulation of digitized information, the blockchain is allowing the borderless flow of digitized value (Freni *et al.*, 2020).

When it comes to the public sector, many governments realize that blockchain is a game-changer and that the sooner they develop facilitating legislation, the greater their jurisdiction's competitive advantage (Sazandrishvili, 2020). Countries, such as Estonia and Sweden are testing e-government solutions, and other government offices have been created to lead digital transformation. Some countries' governments have deployed blockchain projects, which include the Australian Digital Transformation Agency, the US Digital Service and the UK Government Digital Service (Benítez-Martínez *et al.*, 2021).

This paper discusses some utility tokens used to transfer information for the realization of transactions on blockchain such as the use of oracle tokens such as the ChainLink token. In this sense, the primary purpose of this study is to identify the possible efficiency gains in the use of tokenization in expenditure control in the public sector budget execution. The paper presents the use of tokens as an alternative to controlling the physical and financial execution of public property in the second stage of the execution of the expenditure (verification) using the design science research (DSR) methodology to produce an artifact that automates processes and simplifies procedures. We used the Ernst and Young (EY)

OpsChain traceability solution and conducted the validation of the artifact through a focus group of specialist auditors from the public sector. This study extends existing research on blockchain in the accounting literature (e.g. Dai and Vasarhelyi, 2017; Bonson and Bednárová, 2019; Appelbaum and Nehmer, 2020; Sheldon, 2022) examining the creation, transfer and retirement of an asset that exists both in the physical world and as an editable token and proposes directions for future research in this area.

Beyond this introduction, this paper presents a background of related works on blockchain tokenization followed by the research methodology. Subsequently, the design/development and demonstration of the artifact are presented, which includes its description and some detailing of tokens and smart contracts. The evaluation and discussion appear in the penultimate section, which is followed by the conclusion.

2. Tokenization background

A blockchain is a digital public book that chronologically records transactions or events (Kaaniche and Laurent, 2018). Transaction logs are stored in multiple network-linked computers using algorithms to verify and confirm transactions (Schmitz and Leoni, 2019). Blockchain technology operates as an accounting ledger that can record transactions reliably and be used for various purposes (Wieninger *et al.*, 2019). Due to the decentralized nature of the ledger, the greater security is obtained (Rozario and Thomas, 2019), which seems to be the next step in the digital age, has attracted the attention of academics and professionals (Bonson and Bednárová, 2019).

The idea of tokenization is to use smart contracts on a blockchain to create a virtual representation of a particular asset as a token (Mohamed, 2021), to allow them access to all historical transactions and to improve auditing and traceability (Zheng *et al.*, 2020).

Most stablecoins are tokens created on one or several public blockchains managed by a set of smart contracts. The continuous emergence of new assets issued from the interoperability and composability of blockchain, and smart contracts are challenges for accountants and auditors to develop new accounting rules (Carvalho *et al.*, 2022).

Many applications of blockchain are already producing effective implications for companies impacting areas such as governance, transparency, trust, auditing companies with investments in digital assets, real-time auditing and assets' tokenization effect (Silva *et al.*, 2022). For example, Silva *et al.* (2021) produced a proof of concept on tokenization through the creation of a token on the Ethereum Ropsten platform, establishing that the tokenization of currencies, assets, accounts or anything the human mind can conceive is a reality for many companies around the world.

Tokenization allows the creation of fractional rights in a specific asset, as part ownership of real estate can provide customers with the opportunity to invest in a high-value property that they could not otherwise afford. Customers could own a fraction of that property or share the benefits of ownership such as rental income and capital appreciation (Avci and Erzurumlu, 2023). Gupta *et al.* (2020) use security tokens backed by special-purpose vehicles to represent partial ownership of the assets.

2.1 Smart contracts and the functioning of tokens

For Clack *et al.* (2017), a smart contract is an automated and enforceable agreement in computerized systems. Smart contracts are portions of codes where the logic is implemented (Gupta *et al.*, 2020). Stark (2016) highlights the many debates around terminology and its meaning, stating that the term "smart contract" is generally used as a "smart contract code" (a programme code that is stored and executed on a blockchain) or as a complement to or

substitute for legal contracts to run automatically in the blockchain. Blockchain smart contracts can autonomously perform internal control tests and analytical procedures and can be applied to business and audit processes (Rozario and Vasarhelyi, 2018).

Smart contracts, as defined by Grigg (2004), are the same as Ricardian contracts, a digital contract containing all of the terms and clauses as of a regular written contract, but it is readable both by people and by software. The document is digitally signed, and a unique and secure identifier, a cryptographic message digest or hash, is generated over the contents (Simoyama *et al.*, 2017).

An additional advantage of smart contracts in digital ledger technologies is their potential to make verification (nearly) cost less. Consequently, one could expect that without the pressing issue of verification costs, it may become easier to design and offer young firms a much wider menu of financing contracts, where the contractual terms adjust based on incoming and verifiably recorded data (Tinn, 2021). Therefore, considering the tokens are created on top of a blockchain, and governed by smart contracts (Oliveira *et al.*, 2018), reducing costs in transactions by using tokenization seems to be interesting to companies.

After the launch of Bitcoin in early 2009, many other blockchain networks were launched to implement smart contracts. The Ethereum platform as the primary example has a programming language allowing anyone to write smart contracts and applications in a few lines of code (Buterin, 2014).

2.2 Some token types and functions

Because the theme of tokenization is very recent and still requires much research and improvement of knowledge in the market and academia, this paper details the types and characteristics of tokens in this section. Li *et al.* (2019) divide tokens into three types: utility tokens, security tokens and asset-backed tokens, highlighting the following specifications:

- Utility tokens are application coins representing the right to operate the app or access services provided in it. They have no investment attributes.
- A security token is a new kind of crypto token. Its design is based on shares of companies, but it is created on top of a blockchain. The value of a security token is driven by the external value of companies or applications, and it is subject to regulation by the agency to securities exchange.
- Asset-backed tokens represent real-world assets, which could be real estate, art, derivatives markets, non-fungible assets, commodities, services and even identity.

Oliveira *et al.* (2018) present one of the most comprehensive studies on token classification describing both their technical characteristics and business-related aspects. The cited authors described that the arguments for issuing tokens are various, justified by its role in the following dimensions:

- Currency: by acting as transmission of value, unit of account and store of wealth.
- Validation incentive: by ensuring distribution consensus and data consistency.
- Usage incentive: by allowing access or promoting platform usage.
- Tool for accelerating network effects: by incentivizing early adoption.
- Tool for governance: by preventing spamming or providing rights to participate in the platform's development.
- Representation of asset ownership: by encapsulating asset-backed or asset-based property rights.

- Profit-sharing: by conferring its owner the claim to dividends or equivalents.
- Funding instrument: by using the proceeds of a token sale to fund the development team or the community.

Table 1 describes the token features and their attributes (Oliveira *et al.*, 2018).

Mougayar (2017) provides another relevant contribution to capturing the nature of tokens. Mentioned structure was defined around three dimensions of tokens, as depicted in Table 2. The classification focuses on the aspects that involve the business with tokens.

A token's classification depends on its purpose. For tangible assets, there are equity tokens and profit-sharing tokens. From a regulatory perspective, it is relevant to define whether the issuer meets specific regulatory criteria such as those of companies with shares traded on stock exchanges (Sazandrishvili, 2020).

2.3 The Ethereum standard and the fungibility of tokens

The possibility of deploying code copied or initially written in the Ethereum network has caused most tokens created to date to originate from that network. For tokenization, then, it must be followed the proceeding of the Ethereum Request for Comments (ERC), the protocol which became the pattern for tokens created on the Ethereum platform, which are now designated by the acronym ERC followed by the sequential numbering of the respective

Class	Coin/cryptocurrency	Utility token	Tokenized security
Purpose	Asset-backed token	Usage token	Work token
Role	Right	Toll	Currency
Representation	Digital	Physical	Legal
Supply	Fixed	Virtual	Schedule-based
Incentive system	To enter platform	To use platform	To stay long-term
Transactions	Spendable		Non-spendable
Ownership	Tradable		Non-tradable
Fungibility	Fungible		Non-fungible
Layer	Blockchain (native)	Protocol (non-native)	Application (dApp)
Chain	New chain code	New chain, forked code	Issued on top of a protocol

Table 1.
Token classification based on the literature review

Source: Oliveira *et al.* (2018)

Role	Purpose	Features
Right	Bootstrapping engagement	Product usage – voting – governance – product access – contribution – ownership
Value exchange	Economy creation	Work rewards – selling something – buying active/passive work – spending – creating a product
Toll	Skin in the game	Running smart contracts – security deposits – incentive for usage
Function	Enriching user experience	Joining a network – connecting with users – incentive for usage
Currency	Frictionless transactions	Payment unit – transaction unit
Earnings	Distributing benefit	Profit-sharing – benefits sharing – inflation benefits

Table 2.
Guide to crypto tokens usage and value

Source: Mougayar (2017)

request. The ERCs describe provide guidelines that a token should comply with to work correctly on the Ethereum network. Within ERCs, various types of tokens are defined regarding the features of assets such as ERC-20 for divisible assets and ERC-721 for indivisible assets (Wang and Nixon, 2021).

Most existing tokens follow the ERC-20 standard guidelines, and they have some inherent feature that makes one token identical to another in terms of type and value that works similarly to the Ethereum token on the namesake blockchain (Tian *et al.*, 2020). Every ERC-20 is a fungible token. It ensures compatibility between the different Ethereum-based Dapps tokens, thus allowing these tokens to be traded, exchanged or transferred to crypto wallets and facilitating the integration of various projects into the ecosystem (Tian *et al.*, 2020).

One limitation of fungible tokens is that many valuable assets such as artwork and real estate cannot be effectively divided and replaced due to the nature of these assets. Moreover, these assets are typically unique assets, and thus they are better represented by non-fungible tokens (NFTs), that are required to keep track of its ownership (Wang and Nixon, 2021).

An NFT is a cryptographically unique, non-replicable token that can track individual assets' ownership. According to Wang and Nixon (2021), the ERC-721 standard does the same for NFTs as the ERC-20 standard does for the fungible tokens, which allows developers to create a digital asset representation that can be exchanged and tracked on the blockchain. Semi-fungible tokens are a new class of tokens which have the characteristics of FTs and NFTs, enabling them to represent some complex assets or processes through the ERC-1155 (Multi Token Standard), which is another notable variant that offers the potential to create FTs and NFTs in the same contract (Chaleenutthawut *et al.*, 2021). Ethereum developers Radomski *et al.* (2018) created the ERC-1155 as a standard interface for managing multiple token types using a single deployed contract.

2.4 Use of tokens in the public budget process

In computer technology terms, tokens are computer programming code which details what the token represents, which can be a right or any asset. A token may be unrelated to the physical world or tied to a real object, so the token's value varies with the underlying asset. For example, a contract, a company, a car, or any object people can touch and see can be the indexed asset defining a token on a blockchain. Thus, each token depends on the thing to which it is indexed (Li *et al.*, 2019).

Using the tokenization methodology proposed by Li *et al.* (2019), Silva *et al.* (2020) created tokens to represent some budget accounts, demonstrating that the government can monitor and track the phases of forecasting and execution of the public budget on a public blockchain (Abreu *et al.*, 2018). According to International Public Sector Accounting Standard (IPSAS) 11, in construction contracts, such as a binding agreement aimed at building an asset or a set of assets that are interrelated or interdependent, the income and expenses of contracts are recognized to the extent that they are reliably estimated regarding the phase of completion of the contracting activity at the reporting date (Ribeiro, 2015).

In the case of construction, phases are constituted in periodic measurements by the agents appointed by the public administration to monitor the contracts. IPSAS 11 also defines the conditions for disclosure of costs and phases of finishing ongoing contracts. At this point, tokenization can expand the elements of dissemination and advertising and the possibility of popular participation, as seen in the design of tokenization of this paper.

3. Methodology

The theoretical framework underlying the applicability of blockchain technology demonstrates that smart contracts will expand the possibilities of continuous monitoring

(Simoyama *et al.*, 2017) and that smart contracts can be used to autonomously perform internal control tests and analytical procedures (Rozario and Vasarhelyi, 2018), we used DSR to demonstrate the use of tokenization to automate procedures for the physical and financial control and execution of public property. DSR is a scientific approach capable of proposing solutions to practical or knowledge problems (Wieringa, 2009; Rodrigues, 2018) and focuses on the development of:

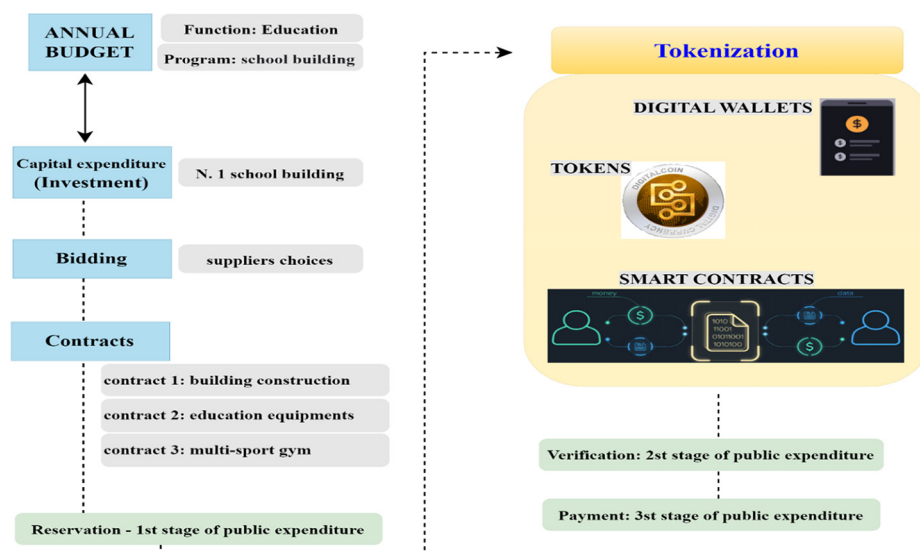
- new knowledge through innovative artifacts; and
- analysis of the use or performance of the artifact (Vaishnavi *et al.*, 2017).

The DSR adopted follows the general approach advocated in the literature (Peffer *et al.*, 2007) based on the cyclical six-step process: problem identification, definition of objectives, design/development, demonstration, evaluation and communication (Ross *et al.*, 2019).

The proposal of the tokenization in this paper is for application after the stage of commitment of expenditure (the first stage of public expenditure execution), as highlighted in Figure 1.

In this paper, the example used is the construction of a school. In the execution of public property, the moment of accounting recognition of the passive (obligation to pay suppliers) occurs in settlement (recognition of the right acquired by the creditor/contractor), with the issuance of the tax document provided and signed by the executor of the contact to attest that the physical execution (PHY) is appropriate according to the contractual terms.

Figure 1 describes the flow of budget expenditure execution, proposing that tokenization is an instrument of control between the first and second stages of expenditure. The proposed scheme is to register in a blockchain (immutable) the actions of the agents involved, aiming to inhibit irregularities, provide improvement of disclosure and add new possibilities for auditing and control because all actions can be checked and monitored in real-time by



Source: Authors' own creation

Figure 1.
Tokenization in the
process of public
expenditure
execution

anyone with permission to access the network, which can be opened to all citizens on a public blockchain depending on regulation.

The creation of tokens follows the methodology proposed by [Silva et al. \(2020\)](#), but this time using the Polygon blockchain. Polygon is a Layer-2 scaling solution created to enable mass adoption of the Ethereum platform. Polygon’s blockchain serves as a commit chain to the Ethereum mainchain on which the network transactions operate adjacent to the main blockchain ([Nailwal, 2021](#)). Polygon is intended to help Ethereum with its scalability. With its focus on providing faster transaction speeds and lower costs than Ethereum, Polygon is a platform to build and run dApps, smart contracts, NFTs and more, but it faces the “blockchain trilemma” of balancing trade-offs between decentralization, security and scalability ([DeNicola, 2023](#)). The smart contracts deployed in the blockchain make the creation and movement of tokens work as a decentralized autonomous organization of each public contract in which the movements of tokens activate the stages of public expenditure.

The development of the artifact for the implementation of the token utility followed the proposal of [Ross et al. \(2019\)](#) in the phases listed by [Peffers et al. \(2007\)](#) as shown in [Figure 2](#).

The next section describes the design, development and demonstration of the artifact. We demonstrated the tokenization using the transactions in the Polygon Testnet blockchain (Mumbai).

The Polygon Mumbai Testnet is a test network created by the Polygon (previously Matic Network) team to provide developers with a sandbox a test environment to try-out their applications on the Polygon network. Like other testnets, Mumbai allows developers to deploy and experiment their applications on the Polygon without having to spend real money ([Alchemy, 2022](#)). Anyone can get MATIC tokens on the Mumbai network, simply by enabling a Metamask wallet, for example, and accessing some Mumbai faucet to receive test tokens (just searching the internet for Mumbai faucet, the place where the wallet address is indicated to receive the MATIC. Since to make any transaction, it is necessary to pay the gas of the trasaction with a small fraction of these tokens available in the connected wallet).

For the evaluation of the artifact, the relevant factors were ranked by auditors in a focus group to define the possibility of improving the quality of the information available in an audited entity that starts to use blockchain tokenization systems. An underlying assumption of this method is that individuals who perform a task can provide valuable

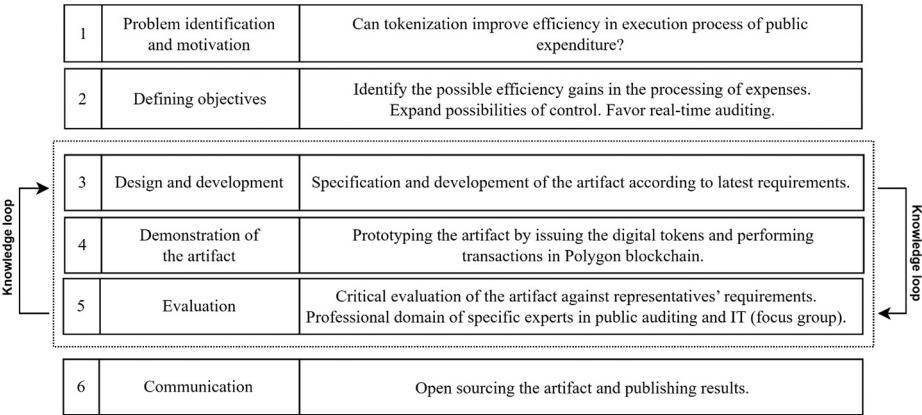


Figure 2.
Design science
research
methodology

Source: Adapted from [Ross et al. \(2019\)](#)

insight into the important factors influencing their ability to achieve a high level of productivity when performing the task (Havelka and Merhout, 2007). Based on Havelka and Merhout (2007), the procedures with the focus group followed six steps as described in Table 3.

In the choice of the team of experts to conduct the focus group, we sought to meet the requirements of knowledge in public-sector auditing, public budget and information technology (IT) and prioritize the possibility of finding people with blockchain knowledge. To this end, we obtained access to auditors from an IT audit department of a Court of Account (CA) in Brazil who are doubly specialized in IT and public auditing in addition to bringing together other specialists from other audit departments who are knowledgeable of governmental budget and accounting. We did this to prioritize a concrete understanding of the design problem among the participants (Myers and Newman, 2007).

The details of the creation of tokens on the blockchain and the use of the OpsChain EY tool are outlined in the next section according to the description and functioning of the artifact in the DSR phases. Finally, as exploratory research, we must highlight that the work relates to new, diffuse concepts involving exploratory discussion of definitions, predictions, opportunities and drawbacks (Frizzo-Barker *et al.*, 2020).

4. Design/development and demonstration

The next section discusses the development of the artifact. In the development phase of a design research project, the tentative direction for an artifact is made concrete through construction and iterative refinement (Kuechler and Vaishnavi, 2008). The demonstration is

Step Description

- 1 Initially, a brief presentation was made to the participants with a general explanation of the study focusing on the use of tokenization after the first stage of the execution of expenditure, notably as to the movement of tokens created to represent each stage and the insertion of metadata to tokens to represent the public agents' actions in the processing of public expense, compared to the ordinary process of which experts have extensive knowledge, without discussing any consequences. The group process for the subsequent collection of the participants' opinions was explained, which began with the presentation of the following question to be answered by the subjects: what factors are relevant to define the quality of the accounting, financial, budgetary or operational information available in the audited entity to issue the auditor's opinion (in other words, what does each participating auditor consider relevant to a good quality of the information provided by the audited entity?)
- 2 We encouraged the auditors to present as many factors as possible that they consider relevant to ensure a high quality of the information available in audited entities, which was recorded in audio for later tabulation
- 3 Subsequently, the facilitator presented supposed positive indicators of the use of tokenization in the process of execution of public expenditure considering the main characteristics of the information maintained in a blockchain, mainly accessibility, transparency, reliability/security, efficiency, the immutability of audited data and the efficiency of data structures
- 4 After a quick description of the factors, participants received a list of these characteristics and their main requirements and were asked to list them in decreasing order of importance if they understood that they would be relevant to improve the quality of the information produced within the audited units and do not indicate them if they considered them irrelevant
- 5 After the individual ranking of the list of factors considered advantageous for the use of tokenization for the quality of the audited entities information, we opened the discussion to clarify the items, distinguish between them and obtain the best consensus in the list
- 6 In the end, we invited the participants to complete a small demographic questionnaire

Source: Authors' elaboration based on Havelka and Merhout (2007)

Table 3.
Focus group steps

viewed in sequence through the tokens generated on the blockchain, which can be widely consulted to access the contracts and wallets in the Polygon block explorer, through the addresses indicated in the endnotes.

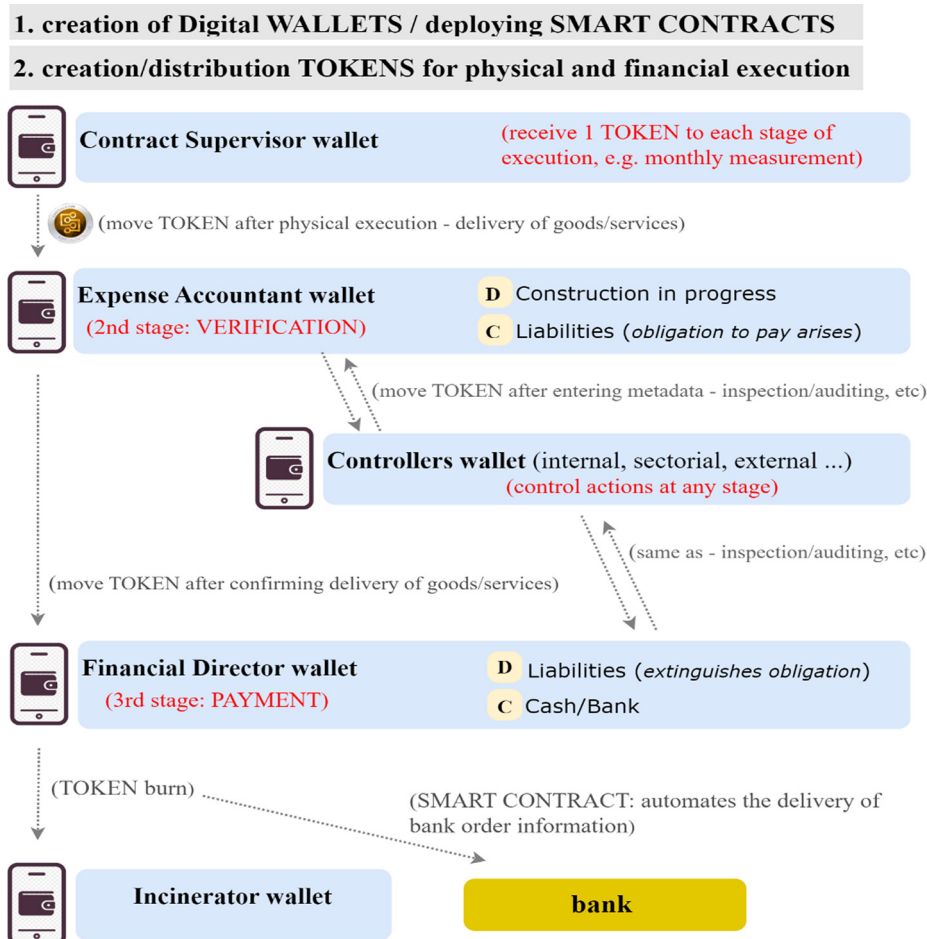
4.1 Artifact description

We present the tokenization process proposed from a conceptual level of abstraction (Ross *et al.*, 2019) and describe the general applicability of the artifact. The design simulates a solution for using tokens as a tool to support the actions of public agents in validating the conference records of the physical and financial execution of a public expense referred to as a school building construction through tokenization on the Polygon blockchain (in the public expense execution second stage – verification). The artifact was produced from a value chain created in the OpsChain EY system, which allows enterprises to access the Ethereum main net and the Polygon proof of stake network.

The EY OpsChain services provide a trusted platform for traceability and transparency within an ecosystem through notarization and tokenization (De Poli, 2022). The EY OpsChain includes the ability to deploy token contracts and then mint, transfer, or burn tokens in addition to allowing the certification of documents to make them immutable through hashing, which means notarization (De Poli, 2022). The EY OpsChain application programming interface services take typical enterprise system outputs, such as serialized parts, batches or products and transform these into unique blockchain tokens carrying history and origin information. Initial services will support non-fungible (ERC-721) tokens with other token types, including ERC-20 and ERC-1155 token standards to follow. Once deployed on a blockchain, these tokens can be transferred, destroyed or updated as needed (Yahoo Inc, 2022).

For our example, we created a value chain in EY OpsChain considering as chain agents the executors and public agents who perform the control function in which tokens are created and moved between the wallets of these agents, which is like what happens with a product supply chain, adapting the use of the EY tool to our case. Wallets represent species of departments that record the path taken by products (tokens). Because the token is present in each wallet, its holder can act as the product owner with permission to insert/change token metadata (an NFT) to record its participation at each stage of the expense execution process, when the token referring for Measurement 1 (e.g. the token minted to represent the production of the first month, as the landing, cleaning and foundation ones) of the construction is available to the contract supervisor (CS), this agent can insert metadata indicating the completion of each part of the construction of school building No. 1).

In this way, each public agent participating in the value chain performs its action by changing the token characteristics existing on the blockchain. The token is moved between the public agents' wallets, and then indicates in the metadata of the token the actions performed, including the auditing professionals, which allows the subsequent tracking, guaranteed by the immutability provided by the functionality of the blockchain. A token can represent anything from the real world (Pietrewicz, 2020), so our study uses this potentiality of tokenization so that each token represents subdivisions of the asset under construction (in this case, a school), in which the data of the production of the public assets, each period of a month, will be recorded in the minted tokens for each period (1, 2, 3 [. . .] n), therefore, tokens were created and whose symbology represents the months of the PHY of the school (PHY1, PHY2, [. . .] PHYn-1, PHYn). For this, it is demonstrated in Figure 5, the example of the token ID 333333, which represents the penultimate phase of the construction of the school (PHY Part n-1).



Source: Authors' own creation

Figure 3.
Tokenization model
to public expense
execution

Our artifact was developed on the Polygon blockchain, which is the network used by EY's OpsChain tool. The Polygon blockchain allows the deployment of smart contracts to automate procedures through wallets and tokens of users, summarized in [Figure 3](#).

It should be noted that tokens can represent, in an analogous way, any other type of asset, be it a building or service, in how many parts you want to represent such as tokens for the walls, the doors and the roof. Tokenization allows the subdivision of assets, in the same way that investors can own portions of a real estate and collectibles such as art. Tokenization in essence allows you to compartmentalize personal data and manage them across different users simply and effectively ([Morrow and Zarrebini, 2019](#)).

The following points detail how digital wallets and the creation/distribution of the tokens work ([Figure 3](#)):

4.1.1 Creation of digital wallets. In the EY OpsChain tool, digital wallets are automatically created for partners operating in the value chain (in our case, the agents of the

expenditure execution), which enables them to act in the tokenization process by creating and moving tokens. The wallets created for the agents defined in Figure 3 are listed in Table 4. The contract creator (0xf0b90E673f296BF76091FB8A1b17A7bce3aABbd8) works as the initial wallet of each public contract through which all tokenization transactions can be verified in the blockchain explorer [1]. If one copies each contract and inserts it into the Polygon blockchain block explorer described in the footnote, all transactions performed in this work can be tracked.

4.1.2 *Creation/distribution of tokens and their operations.* In OpsChain, depending on the permissions set by the value chain authority, partners can tokenize operations or only receive tokens in their wallets, enter information in the form of metadata and transfer tokens after the required actions. For example, an auditor can insert metadata into the token, whose data is written unalterably to the blockchain and guaranteed authenticity by the hash function. In our case, the information on the physical and financial execution of contracts (OFF-chain data from outside the blockchain) depends on the physical action of public agents. They function as living oracles by moving their utility tokens from their wallets, generating ON-chain data for the operation of smart contracts. The example for this paper involved the creation of public expenditure PHY tokens and financial tokens. For simplicity, we highlighted the performance of agents with the movement of only one token, which involved all stages and agents from Figure 3, representing the penultimate phase of physical (PHYN-1 token) execution of a public work (Construction of School No. 1) in the form of the following procedures detailed by partners and expense stages:

4.1.2.1 *Administrative director.* The administrative director (AD) is the legal officer who reports to the CAs and ensures the correct certification of physical and financial execution directly or through the accounting and documentary compliance services subordinated to him. The wallet of this agent is then defined to create and move PHY tokens, ratifying the CS service, if necessary, because he is the hierarchical superior of the management area and responsible for signing the bank order of payment of the expense prepared by the finance/ treasury department. After the supplier issues the tax document/invoice indicating that each phase of construction has been executed, the administrative authority creates the token and moves it to the first agent of the value chain.

4.1.2.2 *Contract supervisor.* The CS is the agent appointed by the administrative authority to oversee the contractual execution, meaning to record that the services were executed as expected. Its function is administrative but does not impact the stages of expense accounting. Therefore, the CSs attest that the service has been properly performed and whose information will be added to the token metadata before moving this to the next agent in the value chain.

4.1.2.3 *Expense accountant – verification – second stage.* The sector responsible for settling the obligation to suppliers must record in the accounting system the payment

Table 4.
Wallets by
organization settings
in the value chain

Users	Partner wallets
Contract supervisor (CS)	0xE2Dc62536805170F4B06FF2845cecf90Ee746C19
Expense accountant (EA) (Verification – second stage)	0x1fC14958048755Cc68337EE9Cfe8780548d57696
Administrative director (AD)	0x763eC780A3a6dF7195b6bE529B70cBeB3607c5e0
Internal control (IC)	0xAa85645C899e58b6820b8E923F71bc8e83dc4217
Court of Account – audit (CA)	0x82398F46b8B7016A8Bac50710545dDB275BF6a32
Financial director (FD) (Bank order – third stage)	0x34a7CbBA388E3D6B3ac66e5c8F9bC2584de4Eac0

Source: Authors' elaboration (wallets linked to Mumbai Testnet – Polygon)

obligation (liabilities) and the existence of the asset (works in progress) due to the execution of the public works. In a fully tokenized system, it will be possible to automate the accounting records from the simple movement of tokens through smart contracts that record the accounting entries. The paper artifact assumes that at every moment, agents perform the procedures in the respective accounting system, which can be overcome in a tokenized environment in which all scanned procedures can be managed by smart contracts.

4.1.2.4 Financial director – payment – third stage. This acts as the chief financial officer, upon receiving the token of the agent who settled the expense, the agent responsible for the financial sector performs the procedures for the payment of the expense. In the action of this agent, the generation of the banking order accounts for the reduction of financial availabilities in return for the termination of the payment obligation (liabilities/suppliers). After performing all procedures and sending the information to the bank, the latter, in turn, proceeds to send the amount of the government debt to the supplier's account. Finally, the financial director (FD) transfer the token to the portfolio in which it will be burned (i.e. the token is moved to the incinerator wallet).

4.1.2.5 Controllers' wallet – internal control, Court of Account. Control wallets (audits and inspections): the agents who perform the control functions can receive the tokens in their wallets at any time. The control agents relate to all other participants in the value chain. In the token creation phase, all control agents have the permission to enter reputational metadata inherent to their understanding of the stage of physical and financial execution of the contract, returning the token to the wallet from which they received it to continue the next stages of the expense execution process. Audit and supervisory processes are generally defined by sampling or scheduled within the financial year. Thus, tokens relating to these agents (as a kind of reputation token) will not necessarily be moved, or this may be done later at the time of the analysis of the accounts. Therefore, reputable tokens may be distributed to various levels of control whether internal to the executing entity itself, internal control units or sectoral or ministerial bodies as well as audit areas and concomitant audits of the courts of accounts or CA units in charge of analysing accountability. By convention, entities or control units do not play a role in the execution of the expense, so their evaluations (conferred reputations) serve only as an indicator of management's efficiency, effectiveness and economics.

4.1.2.6 Tokenization as a digital payment. The tokenizing of this paper has characteristics of representing budget and PHY, but the evolution to a fully tokenized system, tokens of the same nature – as type of cryptocurrencies, which work like money in the blockchain – can also cumulatively represent the means of payment to the vendor. The possibilities allowed by the tokenization are so diverse that it is possible, for example, for tokens to be sent to the supplier's wallet as a means of payment or that the smart contract defines that the shipment of the token to the supplier is accompanied by the shipment of equivalent value in digital currencies to make the payment for the expense, which can be in different types of digital assets such as a central bank digital currency or stable coin.

4.2 Tokens used: Demonstration

The artifact is developed to represent the construction phases of building a school through tokens. The PHY tokens deployed in the Polygon blockchain represent percentages of PHY. Considering that in each execution phase the public agents needs to write new metadata to tokens, PHY tokens are ERC-721. So, they are used to represent the phases of PHY related to the No. 1 School construction (Figure 1). These tokens run as assets that are the result of each stage of construction, and the tokens go through the wallets of each public agent to

Table 5.
Token tracker by
smart contract

indicate the performance by the metadata inserted in each step. The tokens created and their smart contracts generated by the EY OpsChain are listed in [Table 5](#).

All the details of the partners' performances can be viewed almost in real-time on the blockchain as shown in [Figure 4](#).

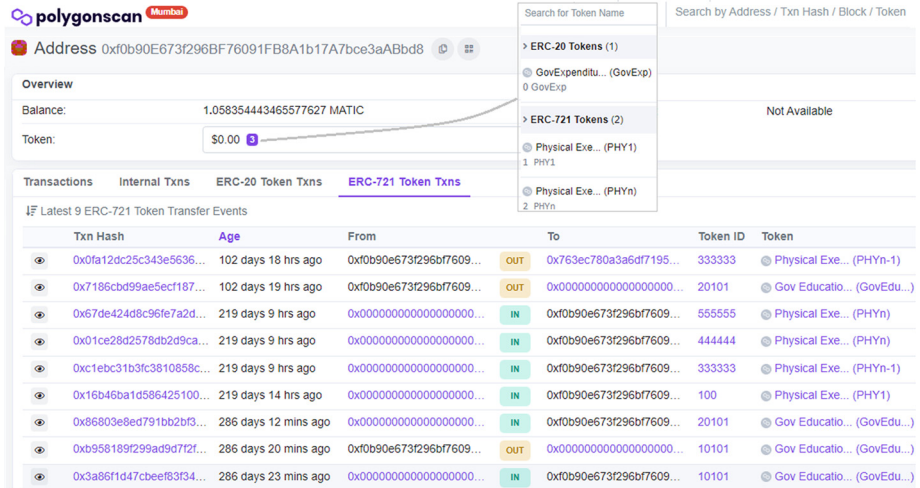
[Figure 4](#) shows the transactions related to the movements of the tokens created and those remaining in the portfolio. The token ID 333333 represents, as an example, the phase n-1 of the construction (token symbol created as PHYn-1) [2]. The movement of this token shows all possible stages in the prototype, from minting and its transfers between the portfolios of public agents to its burning, as seen in [Figure 5](#).

The token ID 333333 (PHY n-1, referring to the penultimate phase of the construction of School No. 1) was moved among the portfolios of all agents (identification of the acronyms in [Table 4](#)) according to the value chain created to represent the execution of the public work. Considering that audit agents, such as the IC and CA can act at any stage, we mention the example of the audit of external control, which received the token shortly after the performance of the FD (payment of the expense). At this time, the token would be burned, so after the auditors' control procedures, the final procedure of the transfer to burning was performed. Before that, auditors inserted text metadata into

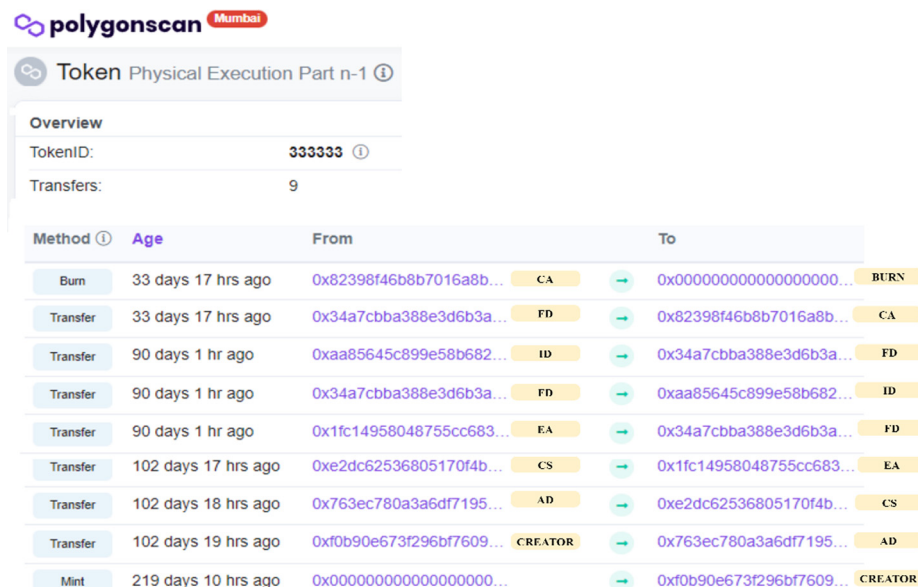
ERC 721 tokens	Contracts
Physical execution Part 1 (PHY1) Part 2 Part 3 [...]	0x8b43103Ba9E6d34c370eD9FCD3D9A25d6cB7748f
Physical execution Part n-1 (PHYn-1) Physical execution Part n (PHYn)	0x3A647D665Bf11A9fE0328d80Ba9043C81053E69e 0x6bb8275528A1B58BA6A61D698720E019D98323f2

Source: Authors' own creation

Figure 4.
Tracking the token
creator contract and
transactions



Source: Authors' own creation



Source: Authors' own creation

Figure 5.
Tracking a token that
represents physical
execution

the system, the hash of which will be written to the blockchain because of audit procedures.

4.3 Evaluation and discussion

To develop the evaluation of the tokenization process (referred to in Figure 2, item 5), we held two focus group sessions. The first discussion session occurred in the IT audit department of a court of auditors in Brazil containing three auditors who specialize in this area and two other experts in accounting and public budgeting to prioritize diverse knowledge in auditing. Considering that the first group contained only male auditors, we held a second session with two female experts, both of whom specialize in auditing, whose additional demographic information is described in Table 6.

To focus on the main aspect of the formation of the focus group, in the first stage, the participants listed the following factors that define good quality of the information provided by the audited entity: trust information, reliable source, traceability, validity, authenticity,

Issue	Description
Education	Five MBAs (accounting, auditing, budget, IT), one master in IT, one PhD student of IT
Years of auditing experience	1 (1–5 y), 3 (5–10 y), 2 (15–20 y), 1 (> 20 y)
Years of TI experience	4 (1–5 y), 2 (5–10 y), 1 (15–20 y)
Self-knowledge in digital technologies	1 insufficient, 3 satisfactory, 2 good, 1 very good
Self-knowledge in blockchain	3 insufficient, 2 satisfactory, 2 good

Source: Authors' own creation

Table 6.
Focus group
demographics

comparability, conference with reality and ability to be reproduced. The characteristics of financial and non-financial information defined by the auditors in the focus group are included in the qualitative characteristics of accounting information defined by the Financial Accounting Standard Board as listed by Hasan *et al.* (2014), which include relevance (predictive value, feedback value and timeliness) and reliability (verifiability, representational faithfulness and neutrality).

In the second stage, the facilitator presented the characteristics provided by information when available in blockchain, namely, accessibility, transparency, reliability/security, efficiency, the immutability of audited data and the efficiency of data structures. The participants then ranked their preferences from 1 to 5 in descending order of importance to form the ranking shown in Table 7.

The descending ranking of relevance of the items shown in Table 7 reveals that six in seven auditors indicated the immutability of the audited data as the most or second more important factor to ensure a good quality of financial and non-financial information for auditing purposes. Given the immutability of blockchain records, the tokenization described in this paper can allow an expansion of control instruments over the executing agents of public spending, possibly increasing efficiency in the public sector.

According to the importance assigned by the auditors to the factors considered advantageous, improving the quality of information using blockchain aligns with potential of tokenization to reduce administrative procedures (Li *et al.*, 2019).

According to the experts, the second-most important point is the reliability and security provided by the information in a tokenized system, which is indicated by the possibility of mitigating risks such as those associated with the existence of fraud resulting from payment for an undelivered physical asset/unexecuted service or even with the induction of overpricing given the greater reliability and immutability of records.

As for reliability and immutability, although blockchains are touted as immutable, not all blockchains are created equally. Hence, the ability to rely on a blockchain depends on several other factors such as the robustness of the consensus mechanism and the reliability of the cryptography involved (Pimentel *et al.*, 2021).

There is evidence that tokenization has the potential to reduce administrative procedures, which become completely transparent and controllable because of the tokenized assets (Li *et al.*, 2019). The focus group validated this characteristic of blockchain and tokenization by ranking transparency as the third most important of the options evaluated.

In this sense, it is worth highlighting Pimentel *et al.* (2021) discussion about “blockchain as a source of verification” highlighting that blockchain positively contributes to verifying whether a transaction occurred and for what amount on what date; however, it does not

Table 7.
Ranking of factors
considered
advantageous by
auditors in the focus
group

	Very high S = 1		Above average S = 2		Average S = 3		Below average S = 4		Very low S = 5		Relevance level
Factors	F	F × S	F	F × S	F	F × S	F	F × S	F	F × S	∑ (F × S)
Immutability	3	3	3	6			1	4			13
Reliability	2	2	2	4	2	6			1	5	17
Transparency	2	2	1	2	2	6	2	8			18
Accessibility			1	2	2	6	3	12	1	5	25
Efficiency							1	4	6	30	34

Source: Authors' own creation

provide an examination of the internal controls that underlie the financial reporting process to prevent or detect fraud and errors. Thus, the literature points to a positive response to this question, but audit activity cannot be supplanted by a blockchain.

The focus group participants said that characteristics of good quality information in the audited entity should be/allow true, timely, current, reliable traceability, veracity of the source, authenticity and comparability, which represent reality and reproducibility. Subsequently, after listening to the properties of the information arranged in blockchain, almost all auditors indicated the immutability as the most important in the scale that forms [Table 7](#). The priority given by the experts to this ability to provide the immutability of the data seems to confirm the auditors' understanding of the good quality of the information for audit purposes.

On the other hand, the experts' indication of the efficiency of data structures as the fifth option with the lowest importance on the scale in [Table 7](#) may indicate the participants' lack of understanding of this potentiality of the use of tokenization to generate good quality information for the auditing purpose. The storage of data in blocks where the recorded data confirm the authenticity of the previous recorded ones can be viewed as the greatest potentiality for the reliability of data in blockchain and elevates tokenization as the future of co-creation ([Kondratyeva et al., 2022](#)).

According to the focus group, all the characteristics listed were relevant; thus, the use of the blockchain system positively contributes to the quality of the information provided by the audited entity. Thus, it is possible to affirm that the evaluation of the artifact can be seen as positive for improving efficiency in the generation of better-quality information by public sector entities mainly due to the immutability and reliability of the data recorded in blockchain.

5. Conclusion

This paper identified the possible efficiency gains from using tokenization in the physical and financial execution of public expenditure using the DSR methodology to produce a construct that allows automation of processes and simplification of procedures simulating control of school building construction through tokenization on the Polygon blockchain.

Through a focus group, we evaluated the possibility of improving the quality of information available in the audited entities that use tokenized systems to represent the actions of public agents in the procedures of each stage of public spending.

Discussions on the topic by experts encourage using tools such as tokenization to improve control and auditability of the physical and financial execution of public spending. The consensus generated among the experts' opinions resulted in a ranking in descending order of importance of the factors that may imply improvements in the quality of the information generated by the public entities which was arranged in the following order: immutability of audited data, reliability/security, transparency, accessibility and efficiency of data structures.

The auditors opined that the greater potential that tokenization and the characteristics of blockchain networks have for the obtention of information in the audited entities is the possibility of the perpetuation of the records, reducing the possibility of deficiency of information by loss of data, whether purposeful or accidental. The auditors considered all listed items relevant concerning the possibility of improved quality of information from the use of tokenization for data records in blockchain.

The consensus obtained in the focus group led to the conclusion that the use of tokenization can be seen as positive to produce better-quality information by public sector entities, notably due to the characteristic of the blockchain to provide immutability and

reliability of the data. This seems to indicate the consequent conclusion that blockchain tokenization might result in other impacts on the budget and financial system linked to public spending such as the possibility of mitigating risks associated with overbilling fraud or payments for improper assets resulting from the effect of tokenization on the control of the physical and financial execution of public goods.

The use of the Polygon blockchain for the implementation of smart contracts and token generation evidenced the possibility of improving the control over the performance of the public manager. New control possibilities can positively affect the sense of supervision over the manager's mindset and the well-known real-time auditing possibilities provided by tokenization, which may be indicated for further investigation.

Future research could expand the scope of the analyses conducted in this study both in terms of a greater scope of experts in the focus group and analysis of new constructs produced using the OpsChain system, which has several other capacities such as notarization, traceability and the capacity to deploy our own programming code of tokenization and smart contracts. As limitations, the lack of effective testing of tokenization with the executing agents of the expense and the managers' opinions about the possibilities of improving the quality of information such as the use of the artifact can be cited.

Notes

1. <https://mumbai.polygonscan.com/address/0xf0b90e673f296bf76091fb8a1b17a7bce3aabbdd8>
2. <https://mumbai.polygonscan.com/token/0x3a647d665bf11a9fe0328d80ba9043c81053e69e?a=333333>

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